

# Galen

## The voice knowledge bank for pharmaceutical manufacturing

Shift handovers carry the knowledge nobody writes down. Galen listens with AssemblyAI, extracts cited knowledge, anchors it to live equipment state, and answers operators by voice.



### Listen

Realtime streaming STT, diarization, keyterms — multilingual plant floor.

### Extract

LLM Gateway turns huddles into typed, cited knowledge entries.

### Answer

Grounded, spoken answers with citations — nothing gets lost.

# Process knowledge evaporates every shift change.

ICH Q10 §1.6.1 requires manufacturers to acquire, analyse, store and disseminate product and process knowledge.

## In people's heads

Knowledge lives in 6 a.m. huddles and handovers — rarely written down.

## Re-discovered

Deviations and near-misses repeat because the fix was never recorded.

## Reconstructed by hand

QA rebuilds "what happened" after the fact, one transcript at a time.

## New-hire churn

New operators repeat mistakes the night shift already solved.

## Multilingual reality

Operators code-switch between English, Luganda, Swahili.

## Audit risk

Knowledge gaps are invisible until an inspection or an incident.

# From huddle to institutional memory — in four steps.

|                                                                                                                                     |                                                                                                                             |                                                                                                                   |                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <p>1 · CAPTURE</p> <p><b>Streaming STT</b></p> <p>AssemblyAI v3 transcribes the huddle live, with speakers and language labels.</p> | <p>2 · EXTRACT</p> <p><b>Cited entries</b></p> <p>LLM Gateway + rule layer produce typed entries that cite exact turns.</p> | <p>3 · ANCHOR</p> <p><b>Live twin</b></p> <p>Metrics update equipment state; threshold breaches raise alerts.</p> | <p>4 · ANSWER</p> <p><b>Cited answers</b></p> <p>Operators ask by voice; get grounded, spoken answers — or book a gap.</p> |
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## Deterministic by design

Every model step has a rule-based fallback, so a known phrase is never lost to a timeout or rate limit.

## Quiet when fine

Only deviations, CAPAs and threshold breaches escalate to a human — approval is explicit.

# Running now.

 Galen

DashboardCaptureAsk

## Plant Overview

### Twin-Lite State

**oeo**  
  
81 %  
2026-09-15T10:51:49.744946

**temperature**  
  
62 °C  
2026-09-15T10:56:08.040766

### Flagged Knowledge Entries

**R2 Temperature Exceeds Limit** deviation\_observation  
R2 is running hot at 63 degrees, feed pulled back.  
By shift\_lead · 2026-09-15T11:34:08.274167 · parsed by llm+rules  

ApproveDismiss

**Issue: Reactor R2 is running hot again this mor...** deviation\_observation  
Reactor R2 is running hot again this morning.  
By shift\_lead · 2026-09-15T10:56:07.935743 · parsed by llm+rules  

ApproveDismiss

3 · Live demo

Record a handover → ask “What do I do when R2 runs hot?”

# AssemblyAI at the core.

## Streaming v3 STT

[universal-3-5-pro](#) live transcription over WebSocket — word-by-word, with speaker diarization and language detection.

## Keyterms prompting

“Reactor R2”, “CAPA”, “out of specification” boosted at the audio layer on a noisy plant floor.

## LLM Gateway

Structured extraction into typed entries with JSON-repair post-processing, in parallel with deterministic rules.

## Grounded Q&A

Retrieval over cited entries; answers audited back to the source turns and twin state.

## Batch + voice notes

Upload a recording in the Ask page and it flows through the same pipeline.

## Browser TTS

Spoken answers, hands-free on the plant floor — no extra backend.

# Vercel frontend, AWS backend, one swap-point storage.

## Vercel (static)

index / capture / ask pages; `config.js` points at the API base with CORS.

## Browser → AssemblyAI

Streaming WebSocket straight from the page; token minted by the backend.

## AWS Lambda + API Gateway

FastAPI via Mangum; async AssemblyAI + LLM calls; 60s budget.

## DynamoDB (galen-kb)

Entries, sessions, twin state, feedback — single-table `pk/sk`.

## Storage abstraction

`agent/repo.py` — SQLite locally, DynamoDB in production, same API.

## Ship it

`scripts/build_lambda.py` +  
`deploy_aws.py` +  
`deploy_vercel.sh`.

# Audit-ready knowledge, by talking.

## ICH Q10 compliant

Acquire, analyse, store, disseminate — every entry cites speaker, turn and twin state.

## Deviation reduction

Root causes from huddles feed the answers the next shift hears.

## Onboarding speed

New operators ask and get grounded answers instead of guessing.

## Multilingual

English / Luganda / Swahili detection, on African plant floors.

## Self-healing knowledge

“No, still stuck” books a gap as a new flagged entry — nothing lost.

## Human in the loop

Approve / Dismiss gate on deviations and CAPA candidates.



THANK YOU

# Galen — talk to your plant.

## Try the demo

<https://galen-kb.vercel.app>

(ask: “What do I do when Reactor R2 runs hot?”)

## Read the code

<https://github.com/MarkNwilliam/galen>

FastAPI + DynamoDB + vanilla JS · MIT

## AssemblyAI

Streaming v3 STT · LLM  
Gateway · diarization

## Backend

AWS Lambda · API Gateway  
· DynamoDB

## Frontend

Vercel · Web Audio · Browser  
TTS

## Domain

ICH Q10 §1.6.1 knowledge  
management